Branching

Mahir Labib Dihan
Lecturer, CSE, BUET

Table of Contents

Introduction to Conditional Statement

The Flow of Execution

Sequential Execution

By default, C programs execute statements one after another, from top to bottom.

However, real life is full of decisions:

- **If** it rains, take an umbrella.
- **If** the traffic light is red, stop; else, go.
- **Switch** the TV channel based on the button pressed.

To handle these, we need **Conditional Statements**.

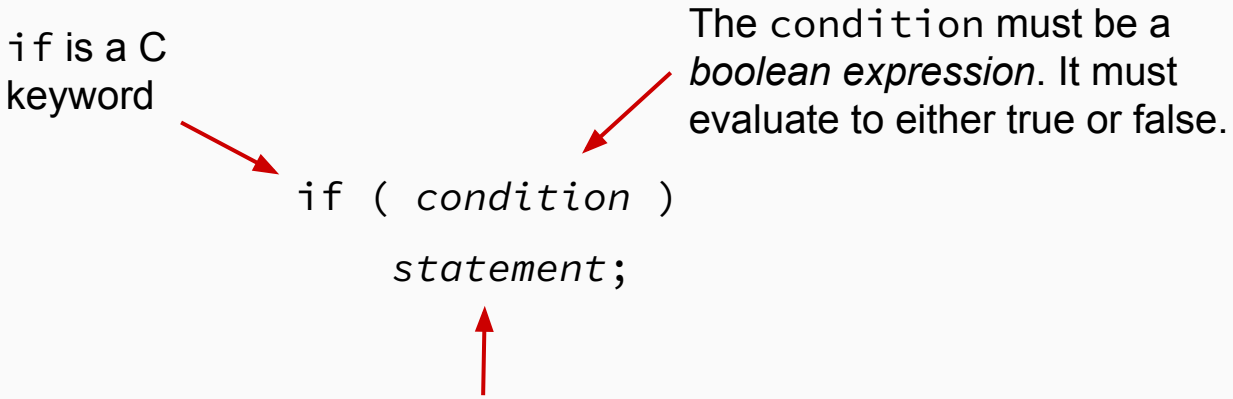
Conditional Statements

- A conditional statement lets us choose which statement will be executed next.
- Therefore they are sometimes called **selection statements**.
- Conditional statements give us the power to make basic decisions.
- The C conditional statements are the:
  - `if` statement
  - `if-else` statement
  - `if-else if-else if-else` ladder
  - `switch-case` statement
  - Conditional operator (`?:`)

The `if` Statement

The if Statement

The if statement has the following syntax:



The if Statement (Example)

Selection structure:

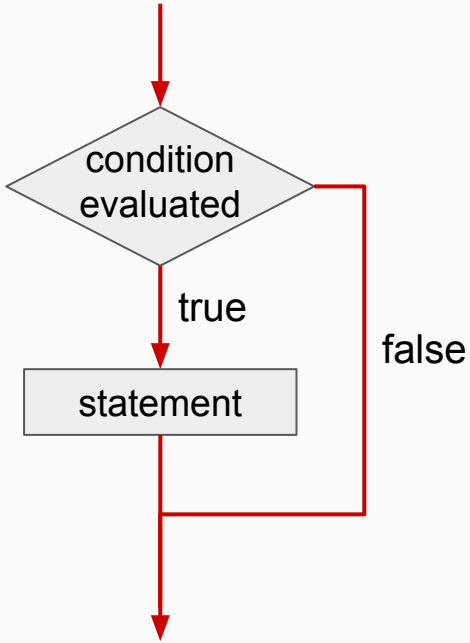
- Used to choose among alternative courses of action
- Pseudocode:

```
If student's mark at least 40
     Print "Passed"
```

Pseudocode statement in C:

```
#include<stdio.h>
int main() {
     int marks;
     printf("Enter your marks: ");
     scanf("%d", &marks);
     if (marks >= 40)
          printf("Passed\n");
     return 0;
}
```

Flowchart: `if` Statement



Relational Operators

A condition often uses one of C's equality operators or relational operators

<	less than	Higher Precedence
>	greater than	
<=	less than or equal to	
>=	greater than or equal to	
==	equal to	Lower Precedence
!=	not equal to	

Note the difference between the equality operator (==) and the assignment operator (=)

The if-else Statement

The if-else Statement

- An else clause can be added to an if statement to make an if-else statement

```
if ( condition )
    statement1;
else
    statement2;
```

- If the `condition` is true, `statement1` is executed;
  if the `condition` is false, `statement2` is executed
- One or the other will be executed, but not both

if-else statement analogy: Y-intersection

The if-else Statement (Example)

Selection structure:

- Used to choose among alternative courses of action
- Pseudocode:

```
If student’s mark at least 40
    Print “Passed”
Otherwise
    Print "Failed"
```

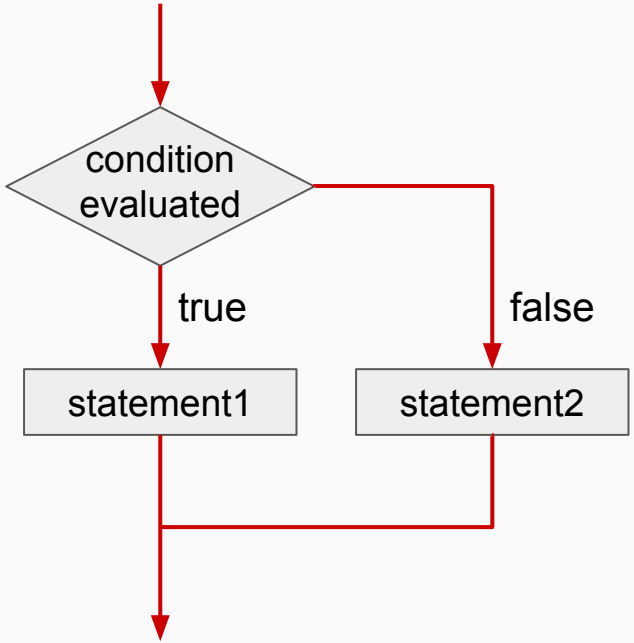
The if-else Statement (Example)

Selection structure:

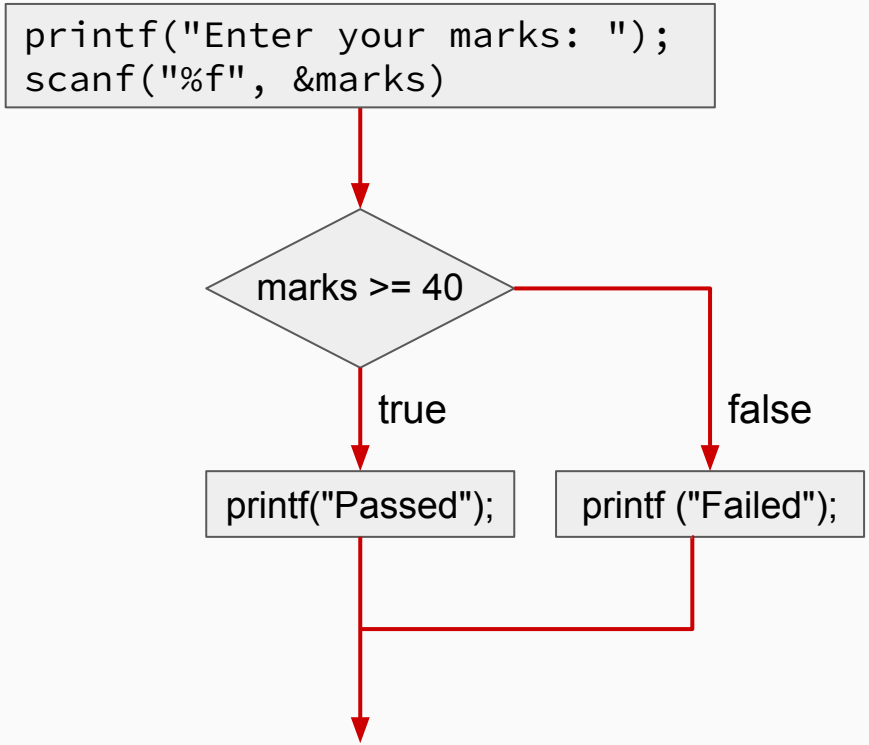
- Used to choose among alternative courses of action
- Pseudocode statement in C:

```
#include<stdio.h>
int main() {
    int marks;
    printf("Enter your marks: ");
    scanf("%d", &marks);
    if (marks >= 40)
        printf("Passed\n");
    else
        printf("Failed\n");
    return 0;
}
```

Flowchart: if-else Statement



Logic of previous example



Values on condition

- Zero (0) → False
- Anything nonzero → TRUE

```
if (40)
    printf("Hi\n");
else
    printf("Bye\n");
// Output: Hi

if (-40)
    printf("Hi\n");
else
    printf("Bye\n");
// Output: Hi
```

Relational Operators

- Zero (0) → False
- Anything nonzero → TRUE

```
a = 0;
if (a)
    printf("Hi\n");
else
    printf("Bye\n");
// Output: Hi

a = 30;
if (a = 0)
    printf("Hi\n");
else
    printf("Bye\n");
// Output: Hi
```

Relational Operators

- FALSE → Zero (0)
- TRUE → One (1)

```
a = 5;
printf("%d", a > 5);
// Output: 0

a = 5;
printf("%d", a == 5);
// Output: 1
```

Quiz

What is the output?

```
a = 6;
b = 5;
c = 2;
if (a > b > c)
    printf("Hi\n");
else
    printf("Bye\n");
```

(a) Compile Error

(b) Hi

(c) Bye ✓

Logical NOT

- The logical NOT operation is also called logical negation or logical complement.
- If some condition a is true, then !a is false; if a is false, then !a is true.
- Logical expressions can be shown using a truth table.

cond	!cond
true	false
false	true

The if Statement (Example)

Selection structure:

- Used to choose among alternative courses of action
- Pseudocode:

```
If student's mark at least 40
     Print "Passed"
```

Pseudocode statement in C:

```c
#include<stdio.h>
int main() {
     int marks;
     printf("Enter your marks: ");
     scanf("%d", &marks);
     if (marks >= 40)
          printf("Passed\n");
     return 0;
}
```

The if Statement (Example)

Selection structure:

- Used to choose among alternative courses of action
- Pseudocode:

```
If student's mark not smaller than 40
     Print "Passed"
```

Pseudocode statement in C:

```
#include<stdio.h>
int main() {
     int marks;
     printf("Enter your marks: ");
     scanf("%d", &marks);
     if (!(marks < 40))
          printf("Passed\n");
     return 0;
}
```

Block Statements

- Several statements can be grouped together into a block statement delimited by braces.
- A block statement can be used wherever a statement is called for in the C syntax rules.

```
if (b == 0)
{
    printf("divide by zero!!\n");
    errorCount++;
}
```

Block Statements

- In an if-else statement, the if portion, or the else portion, or both, could be block statements

```
if (b == 0){
    printf("divide by zero!!");
    errorCount++;
}
else{
    result = a/b;
    printf("Result of division: %d", result);
}
```

Example: Odd or Even

```
#include<stdio.h>
int main() {
    int num;
    printf("Enter an integer: ");
    scanf("%d", &num);

    if ( num % 2 == 0) {
        printf("%d is Even.\n", num);
    } else {
        printf("%d is Odd.\n", num);
    }
    return 0;
}
```

Example: Password Checker

```
int stored_pin = 1234;
int input_pin;

printf("Enter PIN: ");
scanf("%d", &input_pin);

if (input_pin == stored_pin) {
    printf("Access Granted.\n");
    printf("Welcome User.\n");
} else {
    printf("Access Denied.\n");
    printf("Alarm Triggered!\n");
}
```

Exercise

- Write down a program that will take two integers as input and will print the maximum of two.
- Write down a program that will take three integers as input and will print the maximum of three.
- Write down a program that will take three integers as input and will print the second largest of the three.

When we have multiple conditions

`if-else` extension

Vertical Extension

Horizontal Extension

When we have multiple conditions

`if-else` extension

Vertical Extension

Horizontal Extension

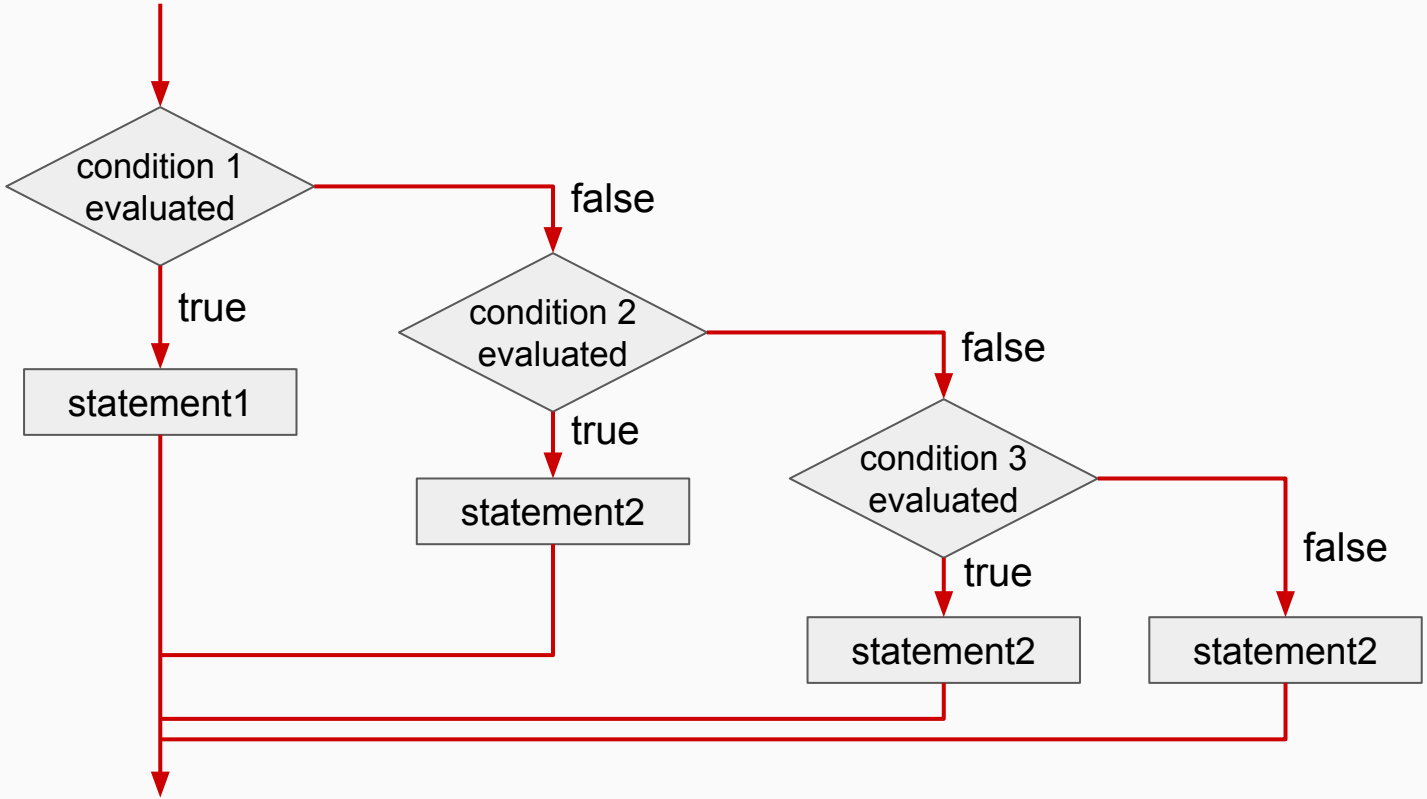
The if-else if-else if–else ladder

- If-else if- else if –else can be used to select from multiple choices:

```
if ( condition1 )
     statement1;
else if ( condition2 )
     statement2;
...
...
else if ( conditionk )
     statementk;
else
     statement;
```

- If the `condition1` is true, `statement1` is executed; if `condition2` is true, `statement2` is executed; and so on

Flowchart: if-else if-else Statement



Example: Grading System

The following chart will be used for a quick grade conversion in C programming language course:

90-100	A
80-89	B
70-79	C
60-69	D
0-59	F

Write down a program that will take a student's mark as input and will convert it to the corresponding letter grade.

Example: Grading System

```
scanf("%d", &m);
if (m >= 90)
    g = 'A';
else if (m >= 80)
    g = 'B';
else if (m >= 70)
    g = 'C';
else if (m >= 60)
    g = 'D';
else
    g = 'F';
printf("Mark = %d Grade= %c", m, g);
```

Example: Grading System

```
scanf("%d", &m);
if (m >= 90)
    g = 'A';
else if (m >= 80)
    g = 'B';
else if (m >= 70)
    g = 'C';
else if (m >= 60)
    g = 'D';
else
    g = 'F';
printf("Mark = %d Grade= %c", m, g);
```

Example: Grading System

```
scanf("%d", &m);
if (m >= 90)
    g = 'A';
if (m >= 80)
    g = 'B';
if (m >= 70)
    g = 'C';
if (m >= 60)
    g = 'D';
else
    g = 'F';
printf("Mark = %d Grade= %c", m, g);
```

Wrong!!

Example: Grading System

```
scanf("%d", &m);
g = 'F';
if (m >= 60)
    g = 'D';
if (m >= 70)
    g = 'C';
if (m >= 80)
    g = 'B';
if (m >= 90)
    g = 'A';
printf("Mark = %d Grade= %c", m, g);
```

Correct but inefficient!!

When we have multiple conditions

`if-else` extension

Vertical Extension

Horizontal Extension

Combining multiple conditions: Logical Operators

- C defines the following logical operators:
  - ! Logical NOT
  - && Logical AND
  - | | Logical OR
- Logical NOT is a unary operator (it operates on one operand). *We have already seen it before.*
- Logical AND and logical OR are binary operators (each operates on two operands).

Logical AND and Logical OR

- The logical AND expression

```
cond1 && cond2
```

  is true if both a and b are true, and false otherwise.

- The logical AND expression

```
cond1 || cond2
```

  is true if both a or b or both are true, and false otherwise.

Logical Operators

- A truth table shows all possible true-false combinations of the terms.
- Since && and || each have two operands, there are four possible combinations of conditions a and b.

`cond1`	`cond2`	`cond1 && cond2`	`cond1 || cond2`
true	true	true	true
true	false	false	true
false	true	false	true
false	false	false	false

Example: Grading System

```
scanf("%d", &m);
if (m >= 90 && m <= 100)
    g = 'A';
else if (m >= 70 && m < 80)
    g = 'C';
else if (m >= 80 && m < 90)
    g = 'B';
else if (m >= 60 && m < 70)
    g = 'D';
else
    g = 'F';
printf("Mark = %d Grade= %c", m, g);
```

Example: Grading System

```
scanf("%d", &m);
if (m >= 90 && m <= 100)
    g = 'A';
if (m >= 70 && m < 80)
    g = 'C';
if (m >= 80 && m < 90)
    g = 'B';
if (m >= 60 && m < 70)
    g = 'D';
else
    g = 'F';
printf("Mark = %d Grade= %c", m, g);
```

Quiz

What is the output?

```
a = 6;
b = 5;
c = 2;
if (a > b && b > c)
    printf("Hi\n");
else
    printf("Bye\n");
```

(a) Compile Error

(b) Hi ✓

(c) Bye

Case Study: The Leap Year Problem

Leap year explained

Problem: Determine if a given year is a Leap Year.

The Math Behind the Rules:

- A solar year is not exactly **365** days long.
- It takes approximately **365.2425** days for Earth to orbit the Sun.
- The extra fraction accumulates over time, so calendar adjustments are needed.
- Leap-year rules help keep the calendar synchronized with Earth's orbit.

Leap year explained

Leap year condition

1, 2, 3, 4, 5, 6, 7, 8, …, 96, 100, 104, …, 200, … ,300, … , 400, …, 500, …, 600, …, 700, …, 800, …, 900, …, 1000… …

- Blue numbers leap year ➤ Multiple of 400
- Red numbers NOT leap year ➤ Multiple of 100 but not of 400
- Green numbers leap year ➤ Multiple of 4 but not of 100
- Black numbers NOT leap year ➤ Not divisible by 4 at all

Solution 1: Vertical Extension (Nested)

Write down a program that will determine whether a year given as input is a leap year or not.

```
if (year % 4 == 0) {
    if (year % 100 == 0) {
        if (year % 400 == 0) {
            printf("Leap Year"); // Divisible by 400
        } else {
            printf("Common Year"); // Divisible by 100 but not 400
        }
    } else {
        printf("Leap Year"); // Divisible by 4 but not 100
    }
} else {
    printf("Common Year"); // Not divisible by 4
}
```

Solution 1: Vertical Extension

Write down a program that will determine whether a year given as input is a leap year or not.

```
if (year % 400 == 0)
    printf("Leap year");
else if (year % 100 == 0)
    printf("Not Leap year");
else if (year % 4 == 0)
    printf("Leap year");
else
    printf("Not a leap year");
```

Solution 2: Horizontal Extension

Leap year condition

1, 2, 3, 4, 5, 6, 7, 8, …, 96, 100, 104, …, 200, … ,300, … , 400, …, 500, …, 600, …, 700, …, 800, …, 900, …, 1000… …

- Blue numbers leap year ➤ Multiple of 400
- Green numbers leap year ➤ Multiple of 4 but not of 100

```
if ( ?? ) {
    printf("Leap Year");
} else {
    printf("Common Year");
}
```

Solution 2: Horizontal Extension

Leap year condition

1, 2, 3, 4, 5, 6, 7, 8, …, 96, 100, 104, …, 200, … ,300, … , 400, …, 500, …, 600, …, 700, …, 800, …, 900, …, 1000… …

- Blue numbers leap year ➤ Multiple of 400
- Green numbers leap year ➤ Multiple of 4 but not of 100

```
if ( blue or green ) {
    printf("Leap Year");
} else {
    printf("Common Year");
}
```

Solution 2: Horizontal Extension

Leap year condition

1, 2, 3, 4, 5, 6, 7, 8, …, 96, 100, 104, …, 200, … ,300, … , 400, …, 500, …, 600, …, 700, …, 800, …, 900, …, 1000… …

- Blue numbers leap year ➤ Multiple of 400
- Green numbers leap year ➤ Multiple of 4 but not of 100

```
if ((Multiple of 400) or (Multiple of 4 but not of 100)) {
    printf("Leap Year");
} else {
    printf("Common Year");
}
```

Solution 2: Horizontal Extension

Leap year condition

1, 2, 3, 4, 5, 6, 7, 8, …, 96, 100, 104, …, 200, … ,300, … , 400, …, 500, …, 600, …, 700, …, 800, …, 900, …, 1000… …

- Blue numbers leap year ➤ Multiple of 400
- Green numbers leap year ➤ Multiple of 4 but not of 100

```
if ((y%400 == 0) || ((y%4 == 0) && (y%100 != 0))) {
    printf("Leap Year");
} else {
    printf("Common Year");
}
```

Nest `if` and `if-else` Ladders

Nested `if`

Putting an `if` inside another `if`

Syntax

```
if (condition1) {
   if (condition2) {
       // Code executes if BOTH 1 and 2 are true
   }
}
```

Use this when the second decision depends on the first one passing

Example: Largest of Three Numbers

```
if (a > b) {
     if (a > c) {
          printf("A is largest");
     } else {
          printf("C is largest");
     }
} else {
     // Here b >= a
     if (b > c) {
          printf("B is largest");
     } else {
          printf("C is largest");
     }
}
```

The Dangling else Problem

Which `if` does the `else` belong to?

Example

```
if (x > 0)
    if (y > 0)
        printf("Both Positive");
else
    printf("???");
```

Rule: An else attaches to the nearest preceding if that doesn't have an else.

Solution: *Always use braces `{}` to be explicit!*

The Art of Indentation

*“Any **fool** can write code that a computer can understand. Good **programmers** write code that humans can understand.”*

- Martin Fowler

What is Indentation?

- Adding spaces or tabs before a statement.
- Used to show which statements belong to a block (`if`, `else`, etc.).
- Makes code easier to read and understand.

Example:

```
if (x > 0) {
    printf("Positive\n");
    printf("Valid Input\n");
}
```

Good vs. Bad Indentation

Bad Code (Flat)

```
if(x>0){
if(y>0){
printf("Q1");
}else{
printf("Q4");
}
}else{
printf("Left");
}
```

Hard to trace.

Good Code (Indented)

```
if (x > 0) {
    if (y > 0) {
        printf("Q1");
    } else {
        printf("Q4");
    }
} else {
    printf("Left");
}
```

Structure is clear.

The Art of Indentation

Why Indent?

- **Readability:** Clearly shows logical structure.
- **Hierarchy:** Visualizes which statements belong to which block.
- **Debugging:** Helps spot missing braces } or logical errors, especially in nested logic.

Note: In C, indentation is ignored by the compiler, but it is critical for you (and whoever is reading your code)!

The Conditional Operator

Conditional Operator (?:)

C has a *conditional operator* that uses a boolean condition to determine which of two expressions is evaluated.

Syntax

```
condition ? expression1 : expression2;
```

- If condition is **`True`**, `expression1` is evaluated.
- If condition is **`False`**, `expression2` is evaluated.e

The value of the entire conditional operator is the value of the selected expression.

Conditional Operator (?:)

The conditional operator is similar to an if-else statement, except that it is an expression that returns a value.

Example

```
larger = ((num1 > num2) ? num1 : num2);
```

- If `num1` is greater than `num2`, then `num1` is assigned to larger;
- Otherwise, `num2` is assigned to larger.

The conditional operator is *ternary* because it requires three operands.

Example: Ternary vs If-Else

Task: Find the maximum of a and b.

Using `if-else`:

```
if (a > b)
     max = a;
else
     max = b;
```

Using Ternary:

```
max = (a > b) ? a : b;
```

Example: Absolute Value

Writing a concise absolute value macro or statement.

```
int num = -10;
int abs_val;

abs_val = (num < 0) ? -num : num;

printf("Absolute value: %d\n", abs_val);
// Output : 10
```

The `switch` Statement

The switch Statement

- The *switch* statement provides another way to decide which statement to execute next.
- The *switch* statement evaluates an expression, then attempts to match the result to one of several possible cases.
- Each case contains a value and a list of statements.
- The flow of control transfers to statement associated with the first case value that matches.

The switch Statement

The general syntax of a `switch` statement is:

```
switch (expression)
{
    case value1:
        // statements
        break;
    case value2:
        // statements
        break;
    default:
        // default statements
}
```

`switch`, `case`, `break` and `default` are keywords

If `expression` matches `value2`, control jumps to here

switch Rules

1. The **expression** must evaluate to an **integer** or **character**. (No float, no string).
2. **`case` values** must be constant literals (e.g., `5`, `'A'`). You cannot use variables like `case x:`.
3. **`break`** prevents "fall-through". Without it, execution continues to the next case.
4. **`default`** is optional (like the final `else`).

The switch Statement: Example

Write down a program using switch-case structure that will take an integer as input and will determine whether the number is odd or even.

```
switch (n % 2)
{
     case 0:
          printf("It is Even");
          break;
     case 1:
          printf("It is ODD");
          break;
}
```

The switch Statement: Example 1

Write down a program using switch structure that will take an integer as input and will determine whether the number is multiple of 3 or not.

```
switch (n % 3)
{
     case 0:
          printf(“It is Multiple of 3”);
          break;
     default:
          printf(“No it’s not”);
          break;
}
```

The switch Statement: Example 1

Write down a program using switch structure that will take an integer as input and will determine whether the number is multiple of 3 or not.

```
switch (n % 3)
{
     case 0:
          printf("It is Multiple of 3");
          break;
     default:
          printf("No it's not");
          break;
}
```

If n is multiple of 3, both will be printed.

The switch Statement: Example 2

Write down a program that will take a character as input and will determine whether it is a vowel or consonant.

```
scanf(“%c”, &ch);
switch (ch)
{
     case 'a':printf("It is Vowel");
              break;
     case 'e':printf("It is Vowel");
              break;
     ……
     case 'u':printf("It is Vowel");
              break;
     default: printf("It is consonant");
}
```

The switch Statement: Fall-through

Write down a program that will take a character as input and will determine whether it is a vowel or consonant.

```
scanf("%c", &ch);
switch (ch)
{
     case 'a':
     case 'e':
     case 'i':
     case 'o':
     case 'u':printf("It is Vowel");
              break;
     default: printf("It is consonant");
}
```

Using *fall-through* feature to group cases.

The switch Statement: Example 3

Write down a program using switch-case structure that will take two integers as input and will determine the maximum of two.

```
switch (a > b)
{
     case 0:
          max = b;
          break;
     case 1:
          max = a;
          break;
}
```

Example 4: Simple Menu

```
int choice ;
printf("1. Start Game\n2. Load Game\n3. Exit\n");
scanf("%d", &choice) ;

switch (choice) {
     case 1:
          printf("Starting...\n");
          break;
     case 2:
          printf("Loading...\n");
          break;
     case 3:
          printf("Goodbye!\n");
          break;
     default:
          printf("Invalid input.\n");
}
```

switch vs if-else

`switch` Case

- Works only with `int` / `char`.
- Checks equality only (`==`).
- Good for menus, state machines.

`if-else` Ladder

- Works with all types (`float`, etc.).
- Checks any condition (`>`, `<`, `&&`, etc.).
- Good for ranges and complex logic.

Problem Solving

A Common Mistake

What is the output of the following code snippet?

```
int x = 10;
if (x = 5) {
    printf("True: %d", x);
} else {
    printf("False: %d", x);
}
```

Output:

```
True: 5
```

Reason: The code uses = (assignment), not == (equality).

1. x is assigned 5.
2. The expression evaluates to 5 (non-zero), which is True.
3. The body executes.

Example: Traffic Light System

Write a switch-case program that prints instructions based on character input:

- `'R'` or `'r'` → “Stop”
- `'Y'` or `'y'` → “Get Ready”
- `'G'` or `'g'` → “Go”
- Other → “Traffic Signal Error”

Solution: Traffic Light System Solution

```
char color ;
scanf (" %c", &color);

switch (color) {
     case 'R':
     case 'r':
          printf("Stop\n");
          break;
     case 'Y':
     case 'y':
          printf("Get Ready\n");
          break ;
     case 'G':
     case 'g':
          printf("Go\n");
          break;
     default:
          printf("Traffic Signal Error\n");
}
```

Example: Grading System

The following chart will be used for a quick grade conversion in C programming language course:

90-100	A
80-89	B
70-79	C
60-69	D
0-59	F

Write down a program that will take a student's mark as input and will convert it to the corresponding letter grade. Solve this using:

1. `if-else if-else` ladder.
2. `switch-case` statement.

Solution A: Grading with if-else

The natural choice for **ranges**.

```
scanf("%d", &m);
if (m >= 90)
    printf("Grade A");
else if (m >= 80)
    printf("Grade B");
else if (m >= 70)
    printf("Grade C");
else if (m >= 60)
    printf("Grade D");
else
    printf("Grade F");
```

Solution B: Grading with switch

Trick: Divide marks by 10 to get the grouping prefix!

```
switch (marks / 10) {
    case 10:
    case 9:
        printf("Grade A"); break;
    case 8:
        printf("Grade B"); break;
    case 7:
        printf("Grade C"); break;
    case 6:
        printf("Grade D"); break;
    default:
        printf("Grade F");
}
```

List of all Keywords in C Language

Keywords in C Programming			
auto	**break**	**case**	char
const	continue	**default**	do
double	**else**	enum	extern
float	for	goto	**if**
int	long	register	return
short	signed	sizeof	static
struct	**switch**	typedef	union
unsigned	void	volatile	while

Summary

1. Use `if` for single conditional execution.
2. Use `if-else` for “this or that” logic.
3. Use `else-if` ladders for ranges and complex multi-condition logic.
4. Use `switch` for checking discrete values (menus, commands).
5. Use `?:` (ternary operator) for simple assignments based on conditions.
6. Always verify indentation and braces to avoid logical errors.

Acknowledgement

- Dr. Ashikur Rahman, Professor, CSE, BUET
- Mohammad Sadat Hossain, Lecturer, CSE, BUET